

5.3.1 Polar Worksheet

1. Consider the polar curve $r = 2 - \sin(\theta)$.

a) Parameterize the polar curve.

We know that

$$\begin{aligned} x &= r \cos(\theta) = [2 - \sin(\theta)] \cos(\theta) \\ y &= r \sin(\theta) = [2 - \sin(\theta)] \sin(\theta) \end{aligned}$$

b) At what value of θ between $[0, 2\pi]$ does the curve reach the point $(x, y) = (0, 1)$?

We need to have

$$\begin{aligned} 0 &= [2 - \sin(\theta)] \cos(\theta) \\ 1 &= [2 - \sin(\theta)] \sin(\theta) \end{aligned}$$

From the first equation we see that either $2 - \sin(\theta) = 0$ (which is impossible) or

$$\cos(\theta) = 0 \rightarrow \theta = \frac{\pi}{2} \text{ or } \theta = \frac{3\pi}{2}.$$

We note that out of these options for θ the only way that the 2nd equation could result in 1 is if $\theta = \frac{\pi}{2}$.

2. Consider the polar equation $r = 2 \sin(3\theta)$.

a) Find a formula for $\frac{dy}{dx}$. No simplification necessary.

We first parametrize the given curve:

$$\begin{aligned} x &= r \cos(\theta) = 2 \sin(3\theta) \cos(\theta) \\ y &= r \sin(\theta) = 2 \sin(3\theta) \sin(\theta) \end{aligned}$$

Now we see that

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{2 \sin(3\theta) \cos(\theta) + 6 \cos(3\theta) \sin(\theta)}{6 \cos(3\theta) \cos(\theta) - 2 \sin(3\theta) \sin(\theta)}$$

- b) Determine the value of θ between $\left(\frac{\pi}{2}, \pi\right)$ when the curve reaches the point $(0, 0)$?

We first need to determine at what value of θ the curve reaches the point $(0, 0)$.

$$\begin{aligned} 0 &= 2 \sin(3\theta) \cos(\theta) \\ \text{We need to have } 0 &= 2 \sin(3\theta) \sin(\theta) \end{aligned}$$

From the system here we see that only way both equations can be satisfied is if $\sin(3\theta) = 0$ (since $\cos(\theta)$ and $\sin(\theta)$ are never equal to 0 at the same time.)

We note that $\sin(3\theta) = 0 \rightarrow 3\theta = 0 + \pi k \rightarrow \theta = \frac{\pi}{3}k$ for some integer k . The only angle here that exists in $\left(\frac{\pi}{2}, \pi\right)$ is $\theta = \frac{2\pi}{3}$.

- c) Determine the slope of the tangent line at the value of θ found in part (b).

We saw from part (b) that $\theta = \frac{2\pi}{3}$. Therefore, we calculate

$$\left. \frac{dy}{dx} \right|_{\theta=\frac{2\pi}{3}} = \frac{2 \sin(2\pi) \cos\left(\frac{2\pi}{3}\right) + 6 \cos(2\pi) \sin\left(\frac{2\pi}{3}\right)}{6 \cos(2\pi) \cos\left(\frac{2\pi}{3}\right) - 2 \sin(2\pi) \sin\left(\frac{2\pi}{3}\right)} = \frac{0 + 6\left(\frac{\sqrt{3}}{2}\right)}{6\left(-\frac{1}{2}\right) - 0} = -\sqrt{3}$$

- d) Determine the *equation* of the tangent line when $\theta = \frac{\pi}{4}$.

We note that the slope of the tangent line is given by

$$\left. \frac{dy}{dx} \right|_{\theta=\frac{\pi}{4}} = \frac{2 \sin\left(\frac{3\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) + 6 \cos\left(\frac{3\pi}{4}\right) \sin\left(\frac{\pi}{4}\right)}{6 \cos\left(\frac{3\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) - 2 \sin\left(\frac{3\pi}{4}\right) \sin\left(\frac{\pi}{4}\right)} = \frac{2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + 6\left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)}{6\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)} = \frac{1-3}{3-1} = \frac{-2}{2} = -1$$

We also note that at $\theta = \frac{\pi}{4}$

$$\begin{aligned} x &= 2 \sin\left(\frac{3\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) = 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = 1 \\ y &= 2 \sin\left(\frac{3\pi}{4}\right) \sin\left(\frac{\pi}{4}\right) = 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = 1 \end{aligned}$$

Therefore we have a tangent line with slope -1 and passing through the point $(1, 1)$.

This line has the equation $y - 1 = -1(x - 1) \rightarrow y = -x + 2$

3. Find the points on the curve given by $r = 3\sin(\theta)$ from $0 \leq \theta \leq 2\pi$ where there are horizontal & vertical tangent lines and singular points.

We note that: $x = f(\theta)\cos(\theta) = 3\sin(\theta)\cos(\theta)$
 $y = f(\theta)\sin(\theta) = 2\sin^2(\theta)$. We must first find when $\frac{dx}{d\theta} = 0$ & $\frac{dy}{d\theta} = 0$.

$$\frac{dx}{d\theta} = 3\cos^2(\theta) - 3\sin^2(\theta)$$

So we need

$$3\cos^2(\theta) - 3\sin^2(\theta) = 0 \rightarrow \cos^2(\theta) = \sin^2(\theta)$$

$$\rightarrow \tan^2(\theta) = 1$$

$$\rightarrow \tan(\theta) = \pm 1$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \& \frac{7\pi}{4}$$

$$\frac{dy}{d\theta} = 4\sin(\theta)\cos(\theta)$$

So we need $4\sin(\theta)\cos(\theta) = 0$

$$\sin(\theta) = 0 \quad \text{or} \quad \cos(\theta) = 0$$

$$\theta = 0, \pi \quad \text{or} \quad \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

Horizontal tangent lines: $\left(\frac{dy}{d\theta} = 0 \& \frac{dx}{d\theta} \neq 0\right)$

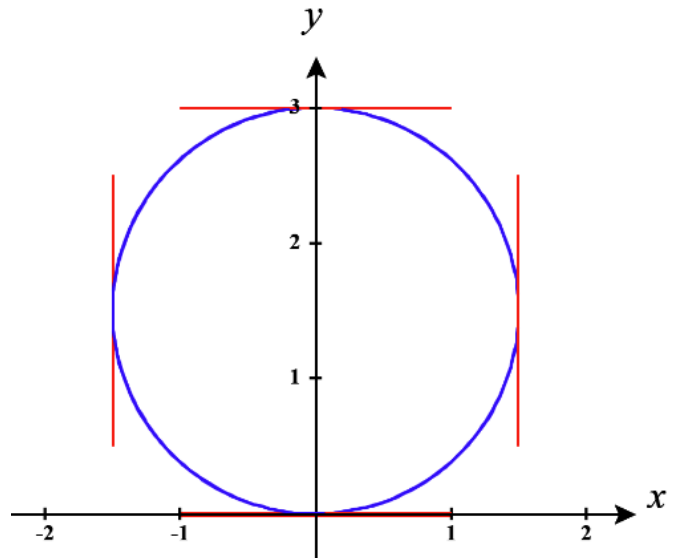
There are horizontal tangents at: $(0,0)$, $\left(3, \frac{\pi}{2}\right)$, $(0, \pi)$, & $\left(-3, \frac{3\pi}{2}\right)$

Vertical tangent line $\left(\frac{dx}{d\theta} = 0 \& \frac{dy}{d\theta} \neq 0\right)$

There are vertical tangents at: $\left(\frac{3\sqrt{2}}{2}, \frac{\pi}{4}\right)$, $\left(\frac{3\sqrt{2}}{2}, \frac{3\pi}{4}\right)$, $\left(\frac{-3\sqrt{2}}{2}, \frac{5\pi}{4}\right)$, & $\left(\frac{-3\sqrt{2}}{2}, \frac{7\pi}{4}\right)$

Singular points $\left(\frac{dy}{d\theta} = 0 \& \frac{dx}{d\theta} = 0\right)$

There are no singular points.



4. Find the points on the curve given by $r = 1 - \sin(\theta)$ from $0 \leq \theta \leq 2\pi$ where there are horizontal & vertical tangent lines and singular points.

We note that: $x = f(\theta)\cos(\theta) = [1 - \sin(\theta)]\cos(\theta)$
 $y = f(\theta)\sin(\theta) = [1 - \sin(\theta)]\sin(\theta)$. We must first find when $\frac{dx}{d\theta} = 0$ & $\frac{dy}{d\theta} = 0$.

$$\begin{aligned}\frac{dx}{d\theta} &= -\cos^2(\theta) - [1 - \sin(\theta)]\sin(\theta) \\ &= -\cos^2(\theta) - \sin(\theta) + \sin^2(\theta) \\ &= -[1 - \sin^2(\theta)] - \sin(\theta) + \sin^2(\theta) \\ &= 2\sin^2(\theta) - \sin(\theta) - 1\end{aligned}$$

So we need

$$2\sin^2(\theta) - \sin(\theta) - 1 = 0 \rightarrow [2\sin(\theta) + 1][\sin(\theta) - 1] = 0$$

$$\sin(\theta) = -\frac{1}{2} \quad \text{or} \quad \sin(\theta) = 1$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \text{or} \quad \theta = \frac{\pi}{2}$$

$$\begin{aligned}\frac{dy}{d\theta} &= -\cos(\theta)\sin(\theta) + [1 - \sin(\theta)]\cos(\theta) \\ &= \cos(\theta) - 2\cos(\theta)\sin(\theta) \\ &= \cos(\theta)[1 - 2\sin(\theta)]\end{aligned}$$

So we need $\cos(\theta)[1 - 2\sin(\theta)] = 0$

$$\cos(\theta) = 0 \quad \text{or} \quad \sin(\theta) = \frac{1}{2}$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2} \quad \text{or} \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Horizontal tangent lines: $\left(\frac{dy}{d\theta} = 0 \text{ \& \; } \frac{dx}{d\theta} \neq 0\right)$

There are horizontal tangents at:

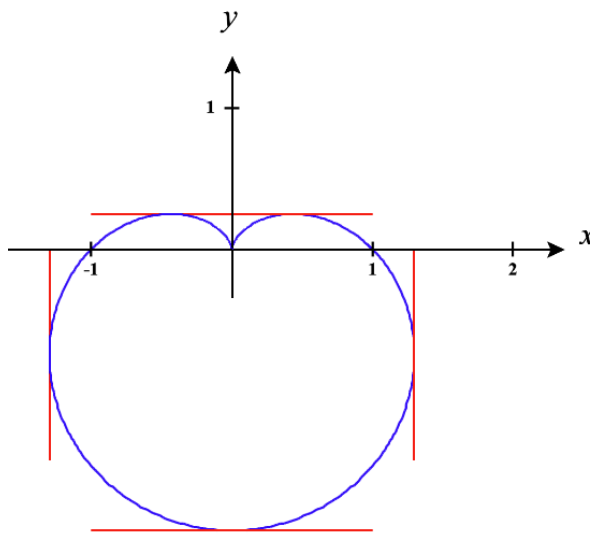
$$\left(\frac{1}{2}, \frac{\pi}{6}\right), \left(\frac{1}{2}, \frac{5\pi}{6}\right), \text{ \& \; } \left(2, \frac{3\pi}{2}\right)$$

Vertical tangent line $\left(\frac{dx}{d\theta} = 0 \text{ \& \; } \frac{dy}{d\theta} \neq 0\right)$

There are vertical tangents at: $\left(\frac{3}{2}, \frac{7\pi}{6}\right) \text{ \& \; } \left(\frac{3}{2}, \frac{11\pi}{6}\right)$

Singular points $\left(\frac{dy}{d\theta} = 0 \text{ \& \; } \frac{dx}{d\theta} = 0\right)$

There is a singular point at: $\left(0, \frac{\pi}{2}\right)$



5. Find the points on the curve given by $r = e^\theta$ from $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ where there are horizontal & vertical tangent lines and singular points.

We note that: $x = f(\theta) \cos(\theta) = e^\theta \cos(\theta)$
 $y = f(\theta) \sin(\theta) = e^\theta \sin(\theta)$. We must first find when $\frac{dx}{d\theta} = 0$ & $\frac{dy}{d\theta} = 0$.

$$\frac{dx}{d\theta} = e^\theta \cos(\theta) - e^\theta \sin(\theta) = e^\theta [\cos(\theta) - \sin(\theta)]$$

$$\frac{dy}{d\theta} = e^\theta \sin(\theta) + e^\theta \cos(\theta) = e^\theta [\sin(\theta) + \cos(\theta)]$$

$$\text{So we need } e^\theta [\cos(\theta) - \sin(\theta)] = 0$$

$$e^\theta = 0 \quad \text{or} \quad \cos(\theta) = \sin(\theta)$$

$$\tan(\theta) = 1$$

$$\text{This never happens or } \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\text{The only value in the given interval is } \theta = \frac{\pi}{4}.$$

$$\text{So we need } e^\theta [\sin(\theta) + \cos(\theta)] = 0$$

$$e^\theta = 0 \quad \text{or} \quad \sin(\theta) = -\cos(\theta)$$

$$\tan(\theta) = -1$$

$$\text{This never happens or } \theta = \frac{3\pi}{4}, -\frac{\pi}{4}$$

$$\text{The only value in the given interval is } \theta = -\frac{\pi}{4}$$

$$\text{Horizontal tangent lines: } \left(\frac{dy}{d\theta} = 0 \text{ \& } \frac{dx}{d\theta} \neq 0 \right)$$

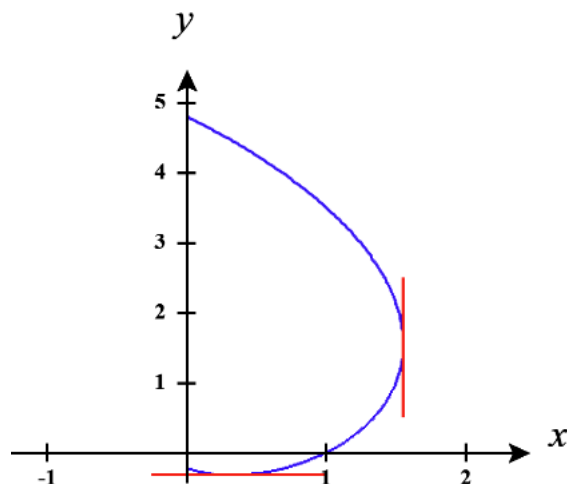
$$\text{There are horizontal tangents at: } \left(e^{-\pi/4}, -\frac{\pi}{4} \right)$$

$$\text{Vertical tangent line } \left(\frac{dx}{d\theta} = 0 \text{ \& } \frac{dy}{d\theta} \neq 0 \right)$$

$$\text{There are vertical tangents at: } \left(e^{\pi/4}, \frac{\pi}{4} \right)$$

$$\text{Singular points } \left(\frac{dy}{d\theta} = 0 \text{ \& } \frac{dx}{d\theta} = 0 \right)$$

There are singular points.



6. Find the points on the curve given by $r = 2\cos^2(\theta)$ from $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ where there are horizontal & vertical tangent lines and singular points.

We note that: $x = f(\theta)\cos(\theta) = 2\cos^3(\theta)$

$y = f(\theta)\sin(\theta) = 2\sin(\theta)\cos^2(\theta)$

We must first find when $\frac{dx}{d\theta} = 0$ & $\frac{dy}{d\theta} = 0$.

$$\frac{dx}{d\theta} = -6\cos^2(\theta)\sin(\theta)$$

So we need $-6\cos^2(\theta)\sin(\theta) = 0$

$\cos(\theta) = 0$ or $\sin(\theta) = 0$

$\theta = -\frac{\pi}{2}, \frac{\pi}{2}$ or $\theta = 0$ (& π , but it isn't in the given range)

$$\begin{aligned}\frac{dy}{d\theta} &= -4\sin^2(\theta)\cos(\theta) + 2\cos^3(\theta) \\ &= 2\cos(\theta)[-2\sin^2(\theta) + \cos^2(\theta)] \\ &= 2\cos(\theta)[-3\sin^2(\theta) + \sin^2(\theta) + \cos^2(\theta)] \\ &= 2\cos(\theta)[-3\sin^2(\theta) + 1]\end{aligned}$$

So we need $2\cos(\theta)[-3\sin^2(\theta) + 1] = 0$

$\cos(\theta) = 0$ or $-3\sin^2(\theta) + 1 = 0$

or $\sin(\theta) = \pm\sqrt{\frac{1}{3}}$

$\theta = -\frac{\pi}{2}, \frac{\pi}{2}$ or $\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right), \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Note here that the angle restrictions on θ allow us to use the inverse sine function.

Horizontal tangent lines: $\left(\frac{dy}{d\theta} = 0 \text{ \& \& } \frac{dx}{d\theta} \neq 0\right)$

There are horizontal tangents at: $\left(\frac{4}{3}, \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$ & $\left(\frac{4}{3}, \sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)$

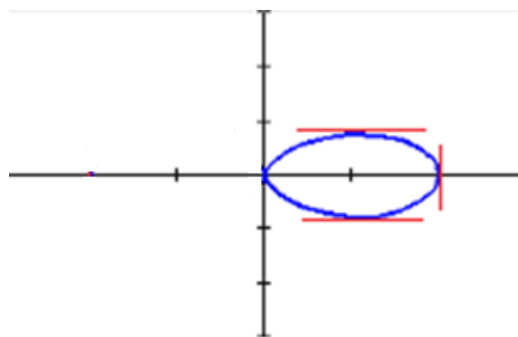
(Note to solve for r you will need to draw a reference triangle.)

Vertical tangent line $\left(\frac{dx}{d\theta} = 0 \text{ \& \& } \frac{dy}{d\theta} \neq 0\right)$

There are vertical tangents at: $(2, 0)$

Singular points $\left(\frac{dy}{d\theta} = 0 \text{ \& \& } \frac{dx}{d\theta} = 0\right)$

There is a singular point at: $\left(0, \frac{\pi}{2}\right)$ & $\left(0, -\frac{\pi}{2}\right)$



7. Determine the exact length of each of the following polar curves.

a) $r = 2 \cos(\theta), 0 \leq \theta \leq \pi$.

We note that the curve is only traced once since no value of r is ever repeated over the given interval and $r = 2 \cos(\theta)$ and $\frac{dr}{d\theta} = -2 \sin(\theta)$ are continuous on $0 \leq \theta \leq \pi$.

Therefore,

$$\begin{aligned} L &= \int_0^{\pi} \sqrt{(2 \cos(\theta))^2 + (-2 \sin(\theta))^2} d\theta = \int_0^{\pi} \sqrt{4 \cos^2(\theta) + 4 \sin^2(\theta)} d\theta = \int_0^{\pi} \sqrt{4} d\theta = \int_0^{\pi} 2 d\theta \\ &= 2\theta \Big|_0^{\pi} = 2\pi \end{aligned}$$

b) $r = 5^{\theta}, 0 \leq \theta \leq 2\pi$

We note that the curve is only traced once over the given interval since no value of r can ever be repeated on the given interval and $r = 5^{\theta}$ and $\frac{dr}{d\theta} = 5^{\theta} \ln(5)$ are continuous on $0 \leq \theta \leq 2\pi$.

Therefore,

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{(5^{\theta})^2 + (5^{\theta} \ln(5))^2} d\theta = \int_0^{2\pi} \sqrt{5^{2\theta} + 5^{2\theta} \ln^2(5)} d\theta = \int_0^{2\pi} \sqrt{5^{2\theta} (1 + \ln^2(5))} d\theta \\ &= \int_0^{2\pi} 5^{\theta} \sqrt{1 + \ln^2(5)} d\theta \\ &= \frac{5^{\theta}}{\ln(5)} \sqrt{1 + \ln^2(5)} \Big|_0^{2\pi} = \frac{5^{2\pi}}{\ln(5)} \sqrt{1 + \ln^2(5)} - \frac{1}{\ln(5)} \sqrt{1 + \ln^2(5)} \end{aligned}$$

c) $r = \theta^2, 0 \leq \theta \leq 2\pi$

We note that the curve is only traced once since no value of r can ever be repeated on the given interval and $r = \theta^2$ and $\frac{dr}{d\theta} = 2\theta$ are continuous on $0 \leq \theta \leq 2\pi$.

Therefore,

$$L = \int_0^{2\pi} \sqrt{(\theta^2)^2 + (2\theta)^2} d\theta = \int_0^{2\pi} \sqrt{\theta^4 + 4\theta^2} d\theta = \int_0^{2\pi} \sqrt{\theta^2(\theta^2 + 4)} d\theta = \int_0^{2\pi} \theta \sqrt{\theta^2 + 4} d\theta$$

Let $u = \theta^2 + 4$ and $du = 2\theta d\theta \rightarrow \frac{1}{2} du = \theta d\theta$. The bounds become $u(0) = 4$ & $u(2\pi) = 4\pi^2 + 4$.

$$\text{So, } \int_0^{2\pi} \theta \sqrt{\theta^2 + 4} d\theta = \int_4^{4\pi^2+4} \frac{1}{2} \sqrt{u} du = \frac{1}{3} u^{3/2} \Big|_4^{4\pi^2+4} = \frac{1}{3} (4\pi^2 + 4)^{3/2} - \frac{1}{3} (4)^{3/2}$$

d) A single tracing of $r = 2[1 + \cos(\theta)]$

We first note that this curve is traced out as θ varies from 0 to 2π .

Also $r = 2[1 + \cos(\theta)]$ and $\frac{dr}{d\theta} = -2\sin(\theta)$ are continuous on $0 \leq \theta \leq 2\pi$.

Therefore,

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{(2[1 + \cos(\theta)])^2 + (-2\sin(\theta))^2} d\theta = \int_0^{2\pi} \sqrt{4(1 + 2\cos(\theta) + \cos^2(\theta)) + 4\sin^2(\theta)} d\theta \\ &= \int_0^{2\pi} \sqrt{8 + 8\cos(\theta)} d\theta = \int_0^{2\pi} \sqrt{8} \sqrt{1 + \cos(\theta)} d\theta \end{aligned}$$

Then note that by the identity

$$\cos^2(x) = \frac{1}{2}[1 + \cos(2x)] \rightarrow \cos(2x) + 1 = 2\cos^2(x) \rightarrow \cos(\theta) + 1 = 2\cos^2\left(\frac{\theta}{2}\right)$$

$$\int_0^{2\pi} \sqrt{8} \sqrt{1 + \cos(\theta)} d\theta = \int_0^{2\pi} \sqrt{8} \sqrt{2\cos^2\left(\frac{\theta}{2}\right)} d\theta = \int_0^{2\pi} 4 \left| \cos\left(\frac{\theta}{2}\right) \right| d\theta.$$

Note that for $0 \leq \theta \leq \frac{\pi}{2}$ & $\frac{3\pi}{2} \leq \theta \leq 2\pi$, $\cos(\theta) \geq 0$ and for $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$, $\cos(\theta) \leq 0$.

Therefore,

$$\begin{aligned} \int_0^{2\pi} 4 \left| \cos\left(\frac{\theta}{2}\right) \right| d\theta &= \int_0^{\pi/2} 4 \cos\left(\frac{\theta}{2}\right) d\theta - \int_{\pi/2}^{3\pi/2} 4 \cos\left(\frac{\theta}{2}\right) d\theta + \int_{3\pi/2}^{2\pi} 4 \cos\left(\frac{\theta}{2}\right) d\theta \\ &= 8 \sin\left(\frac{\theta}{2}\right) \Big|_0^{\pi/2} - 8 \sin\left(\frac{\theta}{2}\right) \Big|_{\pi/2}^{3\pi/2} + 8 \sin\left(\frac{\theta}{2}\right) \Big|_{3\pi/2}^{2\pi} \\ &= \left(8 \left(\frac{\sqrt{2}}{2}\right) - 0\right) - \left(8 \left(\frac{\sqrt{2}}{2}\right) - 8 \left(\frac{\sqrt{2}}{2}\right)\right) + \left(0 - 8 \left(\frac{\sqrt{2}}{2}\right)\right) \\ &= 4\sqrt{2} - 0 - 4\sqrt{2} \\ &= 0 \end{aligned}$$

e) A single tracing of $r = 2\sin(3\theta)$. Do not evaluate this integral.

We first note that this curve is traced out as θ varies from 0 to π .

Also $r = 2\sin(3\theta)$ and $\frac{dr}{d\theta} = 6\cos(3\theta)$ are continuous on

$0 \leq \theta \leq 2\pi$.

Therefore, $\int_0^{\pi} \sqrt{(2\sin(3\theta))^2 + (6\cos(3\theta))^2} d\theta$.

